PRÁCTICA III: PL MÉTODO SIMPLEX

Ejercicio1

$$\text{Max } Z = 2X_1 + 3X_2 + 0X_3 + 0X_4 + 0X_5$$

S.a.:

$$4X_1 + 4X_2 + 1X_3 = 320$$
$$2X_1 + 4X_2 + 1X_4 = 240$$
$$8X_1 + 4X_2 + 1X_5 = 560$$
$$X_j \geq 0$$

	C_j	2	3	0	0	0		
C_i	$A_i \backslash A_j$	A_1	A_2	A_3	A_4	A_5	X_i	X_i/Y_{ij}
0	A_3	4	4	1	0	0	320	80
0	A_4	2	**4**	0	1	0	240	60 →
0	A_5	8	4	0	0	1	560	560/4
Z_j		0	0					
$C_j - Z_j$		2	3 ↑				Z =	0

	C_j	2	3	0	0	0		
C_i	$A_i \backslash A_j$	A_1	A_2	A_3	A_4	A_5	X_i	X_i/Y_{ij}
0	A_3	**2**	0	1	-1	0	80	40 →
3	A_2	1/2	1	0	¼	0	60	120
0	A_5	6	0	0	-1	1	320	320/6
Z_j		3/2			¾			
$C_j - Z_j$		1/2 ↑			3/4		Z = 180	

	C_j	2	3	0	0	0		
C_i	$A_i \backslash A_j$	A_1	A_2	A_3	A_4	A_5	X_i	X_i/Y_{ij}
2	A_1	1	0	½	-1/2	0	40	
3	A_2	0	1	-1/4	½	0	40	
0	A_5	0	0	-3	2	1	80	
Z_j				¼	½			
$C_j - Z_j$				-1/4	-½		Z* = 200	

X^{*T} = (40, 40, 0, 0, 80) **Z* = 200**

Ejercicio 2

a)

$$\text{Max } z = 3x_1 + 9x_2 + 0x_3 + 0x_4$$

s. a

$$1x_1 \quad + 4x_2 \qquad\qquad = \quad 8$$
$$1x_1 \quad + 2x_2 \qquad\qquad = \quad 4$$

$$x_j \quad \geq \quad 0 \qquad j = 1; 2; \ldots; 4$$

			3	9	0	0			x_i/y_{ij}	
	c_i	A_i	A_1	A_2	A_3	A_4		x_i	$(y_{ij}>0)$	
	0	A_3	1	4	1	0		8	2	
	0	A_4	1	2	0	1		4	2	
	z_j		0	0				z = 0		
	$c_j - z_j$		3	9						

			3	9	0	0			x_i/y_{ij}	
	c_i	A_i	A_1	A_2	A_3	A_4		x_i	$(y_{ij}>0)$	
	9	A_2	¼	1	¼	0		2	8	
	0	A_4	½	0	-1/2	1		0	0	
	z_j		9/4		9/4			z = 18		
	$c_j - z_j$		-3/4		9/4					

			3	9	0	0			x_i/y_{ij}	
	c_i	A_i	A_1	A_2	A_3	A_4		x_i	$(y_{ij}>0)$	
	9	A_2	0	1	½	-1/2		2		
	3	A_1	1	0	-1	2		0		
	z_j				3/2	3/2		z = 18		
	$c_j - z_j$				-3/2	-3/2				

$$\boxed{x_1^* = 2;\ x_2^* = 0\ x_3^* = 0;\ x_4^* = 0;\ z^* = 18}$$

Solución degenerada

b) $\text{Min} \quad W = 5\,X_1 + 6\,X_2 - 7\,X_3$

S.a.:

$$X_1 + 5\,X_2 - 3\,X_3 \geq 15$$
$$5\,X_1 - 6\,X_2 + 10\,X_3 \leq 20$$
$$X_1 + X_2 + X_3 = 5$$

$$X_j \geq 0$$

$$\text{Mín } W = 5X_1 + 6X_2 - 7X_3 \quad \text{ó Max } Z = -5X_1 - 6X_2 + 7X_3$$

s.a

$$X_1 + 5X_2 - 3X_3 - 1X_4 + 0X_5 + 1X_6 + 0X_7 = 15$$
$$5X_1 - 6X_2 + 10X_3 + 0X_4 + 1X_5 + 0X_6 + 0X_7 = 20$$
$$X_1 + X_2 + X_3 + 0X_4 + 0X_5 + 0X_6 + 1X_7 = 5$$
$$X_j \geq 0$$

Siendo:

Max $Z = -5X_1 - 6X_2 + 7X_3 + 0X_4 + 0X_5 - MX_6 - MX_7$

	C_j	-5	-6	7	0	0	-M	-M		
C_i	$A_i \backslash A_j$	A_1	A_2	A_3	A_4	A_5	A_6	A_7	X_i	X_i / Y_{ij}
-M	A6	1	5	-3	-1	0	1	0	15	3
0	A5	5	-6	10	0	1	0	0	20	-
-M	A7	1	1	1	0	0	0	1	5	5
Z_j		-2M	-6M	2M	M					
$C_j - Z_j$		-5+2M	-6+6M	7-2M	-M				Z =	-20M

	C_j	-5	-6	7	0	0	-M	-M		
C_i	$A_i \backslash A_j$	A_1	A_2	A_3	A_4	A_5	A_6	A_7	X_i	X_i / Y_{ij}
-6	A2	1/5	1	-3/5	-1/5	0	1/5	0	3	-
0	A5	31/5	0	32/5	-6/5	1	6/5	0	38	95/16
-M	A7	4/5	0	8/5	1/5	0	-1/5	1	2	5/4
Z_j		-6/5-4/5M		18/5-8/5M	6/5-M/5		-6/5+ M/5			
$C_j - Z_j$		- 19/5+4/5M		17/5+8/5M	- 6/5+M/5		6/5-6M/5		Z =	-18-2M

	C_j	-5	-6	7	0	0	-M	-M		
C_i	$A_i \backslash A_j$	A_1	A_2	A_3	A_4	A_5	A_6	A_7	X_i	X_i / Y_{ij}
-6	A2	½	1	0	-1/8	0	1/8	3/8	15/4	
0	A5	3	0	0	-2	1	2	-4	30	
7	A3	½	0	1	1/8	0	-1/8	5/8	5/4	
Z_j		1/2			13/8		-13/8	17/8		
$C_j - Z_j$		-11/2			-13/8		-M+13/8	-M-17/8	Z* = -55/4 W*= 55/4	

$X^{*T} = (0, 15/4, 5/4, 0, 30)$ $W^* = 55/4$

b) Resolver utilizando el Método de las dos Fases

FASE I

Programa auxiliar:

$\text{Min } \Phi = X6 + X7$ o $\text{Max } \zeta = -X6 - X7$

s.a

$$
\begin{aligned}
X_1 + 5X_2 - 3X_3 - 1X_4 + 0X_5 + 1X_6 + 0X_7 &= 15\\
5X_1 - 6X_2 + 10X_3 + 0X_4 + 1X_5 + 0X_6 + 0X_7 &= 20\\
X_1 + X_2 + X_3 + 0X_4 + 0X_5 + 0X_6 + 1X_7 &= 5
\end{aligned}
$$

$$X_j \geq 0$$

	C_j	0	0	0	0	0	-1	-1		
C_i	$A_i \backslash A_j$	A_1	A_2	A_3	A_4	A_5	A_6	A_7	X_i	X_i/Y_{ij}
-1	A6	1	5	-3	-1	0	1	0	15	3 →
0	A5	5	-6	10	0	1	0	0	20	-
-1	A7	1	1	1	0	0	0	1	5	5
Z_j		-2	-6	2	1					
$C_j - Z_j$		2	6 ↑	-2	-1				$\zeta = -20$	

	C_j	0	0	0	0	0	-1	-1		
C_i	$A_i \backslash A_j$	A_1	A_2	A_3	A_4	A_5	A_6	A_7	X_i	X_i/Y_{ij}
0	A2	1/5	1	-3/5	-1/5	0	1/5	0	3	-
0	A5	31/5	0	32/5	-6/5	1	6/5	0	38	95/16
-1	A7	4/5	0	8/5	1/5	0	-1/5	1	2	5/4 →
Z_j		-4/5		-8/5	-1/5		1/5			
$C_j - Zj$		4/5		8/5 ↑	1/5		-6/5		$\zeta = -2$	

	C_j	0	0	0	0	0	-1	-1		
C_i	$A_i \backslash A_j$	A_1	A_2	A_3	A_4	A_5	A_6	A_7	X_i	X_i / Y_{ij}
0	A2	½	1	0	-1/8	0	1/8	3/8	15/4	
0	A5	3	0	0	-2	1	2	-4	30	
0	A3	½	0	1	1/8	0	-1/8	5/8	5/4	
Z_j		0			0		0	0		
C_j- Zj		0			0		-1	-1	ζ *= 0	

FASE II

PL Original: **Max $Z = -5X_1 - 6X_2 + 7X_3$**

	C_j	-5	-6	7	0	0	-1	-1		
C_i	$A_i \backslash A_j$	A_1	A_2	A_3	A_4	A_5	A_6	A_7	X_i	X_i / Y_{ij}
-6	A2	½	1	0	-1/8	0			15/4	
0	A5	3	0	0	-2	1			30	
7	A3	½	0	1	1/8	0			5/4	
Z_j		1/2			13/8					
C_j- Z_j		-11/2			-13/8				Z* = -55/4 W*= 55/4	

X^{*T} = (0, 15/4, 5/4, 0, 30) **W* = 55/4**

c) Max $Z = 3X_1 + 2X_2$

S.a.:

$$2X_1 + X_2 \le 2$$
$$3X_1 + 4X_2 \ge 12$$
$$X_j \ge 0$$

Llevarlo a la forma estándar

Máx $Z = 3X_1 + 2X_2 + 0X_3 + 0X_4$

S.a.:

$$2X_1 + X_2 + 1X_3 + 0X_4 = 2$$
$$3X_1 + 4X_2 + 0X_3 - 1X_4 = 12$$
$$X_j \ge 0$$

b) Resolverlo por tablas (Método de las dos fases)

FASE I

Programa auxiliar:

$$\text{Max } \zeta = -X5$$

S.a.

$$2X_1 + X_2 + 1X_3 + 0X_4 = 2$$

$$3X_1 + 4X_2 + 0X_3 - 1X_4 + X_5 = 12$$

$$X_j \geq 0$$

	C_j	0	0	0	0	-1		
C_i	$A_i \backslash A_j$	A_1	A_2	A_3	A_4	A_5	X_i	X_i/Y_{ij}
0	A3	2	1	1	0	0	2	2 →
-1	A5	3	4	0	-1	1	12	3
Z_j		-3	-4		1			
$C_j - Z_j$		3	4 ↑		-1		$\zeta = -12$	

	C_j	0	0	0	0	-1		
C_i	$A_i \backslash A_j$	A_1	A_2	A_3	A_4	A_5	X_i	X_i/Y_{ij}
0	A2	2	1	1	0	0	2	
-1	A5	-5	0	-4	-1	1	4	
Z_j		5		4	1			
$C_j - Z_j$		-5		-4	-1		$\zeta = -4$	

Se alcanzó la condición de **óptimo con** $\zeta \neq 0$

y la variable ficticia: **$X5 = 4 \neq 0$**

Luego, el PL original **NO tiene Solución Factible**

d) Solución en el infinito(lindo)

e) $\text{Mín } W = 24X_1 + 36X_2 + 24X_3 + 36X_4$

S.a

$$6X_1 + 4X_2 = 100$$

$$2X_3 + 3X_4 = 24$$

$$X_j \geq 0$$

Por penalización

$$\text{Mín } W = 24\,X_1 + 36\,X_2 + 24\,X_3 + 36\,X_4 + M\,X_5 + M\,x_6$$

S.a:

$$6\,X_1 + 4\,X_2 + 0\,X_3 + 0\,X_4 + 1\,X_5 + 0\,X_6 = 100$$
$$0\,X_1 + 0\,X_2 + 2\,X_3 + 3\,X_4 + 0\,X_5 + 1\,X_6 = 24$$
$$X_j \geq 0$$

	C_j	24	36	24	36	M	M		
C_i	$A_i \backslash A_j$	A_1	A_2	A_3	A_4	A_5	A_6	X_i	X_i/Y_{ij}
M	A5	6	4	0	0	1	0	100	100/6 →
M	A6	0	0	2	3	0	1	24	-
Z_j		6M	4M	2M	3M				
$C_j - Z_j$		-6M+24 ↑	-4M+36	-2M+24	-3M+36			W =	124M

	C_j	24	36	24	36	M	M		
C_i	$A_i \backslash A_j$	A_1	A_2	A_3	A_4	A_5	A_6	X_i	X_i/Y_{ij}
24	A1	1	4/6	0	0	1/6	0	100/6	-
M	A6	0	0	2	3	0	1	24	8 →
Z_j			16	2M	3M	4			
$C_j - Z_j$			20	-2M+24	-3M+36 ↑	-4+M		W =	400+24M

	C_j	24	36	24	36	M	M		
C_i	$A_i \backslash A_j$	A_1	A_2	A_3	A_4	A_5	A_6	X_i	X_i/Y_{ij}
24	A1	1	4/6	0	0	1/6	0	100/6	
36	A4	0	0	2/3	1	0	1/3	8	
Z_j			16	24		4	12		
$C_j - Z_j$			-20	0		-4+M	-12+M	W*=	688

$X^{*T} = (\,100/6,\ 0,\ 0,\ 8)$ $\qquad\qquad$ $W^* = 688$

<u>Solución alternativa:</u>

Como C_3- Z_3 = 0 siendo X_3 NO básica, la solución alternativa se obtiene haciendo ingresar a A_3 a la base:

	C_j	24	36	24	36	M	M		
C_i	$A_i \backslash A_j$	A_1	A_2	A_3	A_4	A_5	A_6	X_i	X_i/Y_{ij}
24	A1	1	4/6	0	0	1/6	0	100/6	
24	A3	0	0	1	3/2	0	1/2	12	
Z_j			16		36	4	12		
C_j- Z_j			-20		0	-4+M	-12+M	W*=	688

X^{*T} = (100/6, 0, 12, 0) **W* = 688**